

Understanding the Mediation Formula: Intuition and Derivation with Interaction

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The mediation formula is a fundamental result in causal inference that allows us to express a specific counterfactual quantity in terms of observable data. The formula is given by:

$$\mathbb{E}[Y^{a,M^{a^*}} | C = c] = \sum_m \mathbb{E}[Y | A = a, M = m, C = c] P(M = m | A = a^*, C = c)$$

This formula is a cornerstone of modern mediation analysis, often called the **Mediation Formula** or the **Mediation G-Formula**. It provides a non-parametric way to *identify* a very specific and important counterfactual quantity from observed data, *assuming* a set of critical assumptions hold.

The quantity on the left-hand side, $\mathbb{E}[Y^{a,M^{a^*}} | C = c]$, is the key building block for defining **Natural Direct Effects** (NDE) and **Natural Indirect Effects** (NIE).

Let's break this down in detail, starting with the causal question on the left and then moving to the statistical calculation on the right.

1. The Causal Question: The Left-Hand Side (LHS)

The term $\mathbb{E}[Y^{a,M^{a^*}} | C = c]$ represents a “nested” or “cross-world” potential outcome.

- **Notation:**

- A : The exposure (e.g., a = treatment, a^* = control).
- M : The mediator (e.g., a biomarker, a health behavior).
- Y : The outcome (e.g., disease status, recovery).
- C : A set of baseline (pre-exposure) confounders.
- Y^a : The potential outcome Y if, possibly contrary to fact, A were set to a .
- M^{a^*} : The potential mediator value M if, possibly contrary to fact, A were set to a^* .

- **The Full Term:** $\mathbb{E}[Y^{a,M^{a^*}} | C = c]$ This asks a sophisticated causal question: > “Within a specific stratum of confounders $C = c$, what would be the average outcome Y if we intervened to set the exposure to level a , but simultaneously intervened to set the mediator M to the value it *would have naturally taken* if the exposure had been set to a^* ?”

This is a “cross-world” counterfactual because it combines two different hypothetical worlds:

1. The exposure is set to a .
2. The mediator is set to a value derived from a world where the exposure was a^* .

Practical Example (Public Health): Imagine A is a new, mandatory worksite wellness program ($a = 1$) vs. no program ($a^* = 0$). M is the level of physical activity, and Y is 5-year cardiovascular disease (CVD) risk. C are baseline demographics and health status.

The formula $\mathbb{E}[Y^{a=1,M^{a^*=0}} | C = c]$ asks:

Note

What would the average CVD risk be if we *implemented the program* ($A = 1$), but hypothetically, everyone maintained the *same level of physical activity* (M) that they *would have had if there were no program* ($A = 0$)?

This quantity is precisely what we need to isolate the **Natural Direct Effect (NDE)**. The NDE compares this to the risk under no program:

$$\text{NDE} = \mathbb{E}[Y^{a=1,M^{a^*=0}} | C = c] - \mathbb{E}[Y^{a^*=0,M^{a^*=0}} | C = c]$$

This isolates the effect of the program on CVD risk that does *not* operate through physical activity.

2. The Statistical Answer: The Right-Hand Side (RHS)

The left side is the unobservable *question*. The right side is the *answer* we can calculate from observable data, assuming identification.

$$\sum_m \mathbb{E}[Y | A = a, M = m, C = c] P(M = m | A = a^*, C = c)$$

RHS: The Identification Formula

This formula is an application of **standardization**. Let’s break down its two components.

Term 1: $\mathbb{E}[Y | A = a, M = m, C = c]$

- **What it is:** The observed conditional expectation of the outcome.
- **Intuitive Meaning:** “Among individuals who *actually received* exposure $A = a$, what is the average outcome Y for those who had a specific mediator value $M = m$ (within stratum $C = c$)?”
- **How we get it:** This is estimated directly from our data. We can fit a regression model for the outcome, $\mathbb{E}[Y | A, M, C]$, and plug in the values $A = a$ and $M = m$.
- **Example (cont.):** “In our study, what is the average CVD risk among people who *were in the wellness program* ($A = 1$) and *also* had a physical activity level of m (e.g., $m=\text{low}$)?” This term represents the direct pathway from A to Y and the M to Y pathway, conditional on A, C .

Term 2: $P(M = M | A = a^*, C = c)$

- **What it is:** The observed conditional probability (or density) of the mediator.
- **Intuitive Meaning:** “What is the *distribution* of the mediator M among individuals who *actually received* the control exposure $A = a^*$ (within stratum $C = c$)?”
- **How we get it:** This is also estimated from data. We fit a model for the mediator, $P(M | A, C)$, and plug in $A = a^*$.
- **Example (cont.):** “In our study, what *proportion* of people *not in the wellness program* ($A = 0$) had a low physical activity level ($M=\text{low}$)?” This term represents the natural distribution of the mediator under the control condition.

3. What Does “Integrating out the mediator” Mean?

This is the most important part of your question and is the key to the intuition.

The $\sum_m$ (or $\int \dots dm$ for a continuous mediator) is the “integration.” It means we are calculating a **weighted average**.

“Integrating out the mediator” means we are performing standardization.

Here is the step-by-step intuition:

1. **Goal:** We want the average outcome under exposure $A = a$, but with the mediator distribution from $A = a^*$.
2. **Step 1: Get the outcome relationship under $A = a$:** We use Term 1: $\mathbb{E}[Y | A = a, M = m, C = c]$. This gives us a *list* of average outcomes for every possible value of m , *all* in the group that received exposure $A = a$.
 - *Example:*
 - Avg. CVD risk for (Program, $M=\text{low}$) = 15%

- Avg. CVD risk for (Program, $M=\text{medium}$) = 10%
- Avg. CVD risk for (Program, $M=\text{high}$) = 5%

3. **Step 2: Get the mediator distribution under $A = a^*$:** We use Term 2: $P(M = m \mid A = a^*, C = c)$.

This gives us a *list* of weights (probabilities).

- *Example:* In the *No Program* group:
 - Proportion with $M=\text{low}$ = 60%
 - Proportion with $M=\text{medium}$ = 30%
 - Proportion with $M=\text{high}$ = 10%

4. **Step 3: Standardize (Integrate/Sum):** We combine them. We “average” the outcomes from Step 1 using the “weights” from Step 2.

$$\text{Result} = (15\% \times 60\%) + (10\% \times 30\%) + (5\% \times 10\%)$$

$$\text{Result} = (0.15 \times 0.60) + (0.10 \times 0.30) + (0.05 \times 0.10)$$

$$\text{Result} = 0.09 + 0.03 + 0.005 = 0.125 \text{ or } 12.5\%$$

This result, 12.5%, is the value of $E[Y^{a=1, M^{a^*=0}} \mid C = c]$. It’s a *standardized* outcome. It tells us the average CVD risk we would expect in the Program group (12.5%) *if* they had the same (less healthy) physical activity distribution as the No Program group.

“Integrating out” simply means we are no longer looking at M -specific outcomes; we have collapsed them into a single, standardized average.

4. The Assumptions Are Paramount

This formula, this entire identification strategy, is *not* just a statistical manipulation. It only works—meaning the RHS truly equals the LHS—if four key identification assumptions hold (conditional on C):

1. No unmeasured Exposure-Outcome confounding.
2. No unmeasured Mediator-Outcome confounding.
3. No unmeasured Exposure-Mediator confounding.
4. No Mediator-Outcome confounder is itself affected by the Exposure.

This last assumption is the most challenging. It means there can't be a variable L that is caused by A , and which in turn causes both M and Y . If such a variable exists, we cannot use this simple formula and must turn to more advanced methods (like MSMs for mediation or sequential g-formula).

Next Steps

Now that we have the theoretical formula, the next logical step is to think about how we would practically estimate this from a dataset. This involves:

1. Fitting a statistical model (e.g., logistic regression, linear regression) for the outcome Y given A, M , and C .
2. Fitting a statistical model (e.g., logistic regression, linear regression, multinomial) for the mediator M given A and C .
3. Using the procedure I described (sometimes called “parametrically-adjusted” estimation or “g-computation”) to calculate the standardized means and compute the Natural Direct and Indirect Effects.

Suggested Follow-up Prompts

1. How would we write the full formulas for the **Natural Direct Effect (NDE)** and **Natural Indirect Effect (NIE)** using this notation?
2. Can you draw a DAG that *violates* the fourth assumption (no exposure-induced M-Y confounder) and explain why this formula would fail?
3. How would we use R (e.g., with the `mediation` package or by hand) to estimate the two terms in this formula and compute the NDE?

Exposure-Mediator Interaction

When an exposure-mediator interaction exists, the effect of the mediator on the outcome *depends on the level of the exposure*. The counterfactual approach allows us to correctly parse this.

Let's walk through the derivation using the regression models you've proposed.

1. Define the Statistical Models

We are in a setting with baseline confounders C , a binary exposure $A \in \{0, 1\}$, a continuous mediator M , and a continuous outcome Y .

Model 1: The Mediator Model We assume a linear model for M , conditional on A and C :

$$\mathbb{E}[M \mid A = a, C = c] = \beta_0 + \beta_1 a + \beta_2' c$$

For the derivation, we will also assume the mediator's distribution is normal, $M \mid A = a, C = c \sim \mathcal{N}(\beta_0 + \beta_1 a + \beta_2' c, \sigma_m^2)$. The β_1 coefficient represents the effect of A on M .

Model 2: The Outcome Model We assume a linear model for Y , conditional on A , M , and C . Crucially, we include the $A \times M$ interaction term.

$$\mathbb{E}[Y \mid A = a, M = m, C = c] = \theta_0 + \theta_1 a + \theta_2 m + \theta_3' c + \theta_4 (a \times m)$$

Here:

- θ_1 is the “direct” effect of A on Y , when $M = 0$.
- θ_2 is the effect of M on Y , when $A = 0$.
- θ_4 is the interaction term. It tells us *how much the effect of M on Y changes* when A changes from 0 to 1. The total effect of M on Y when $A = 1$ is $(\theta_2 + \theta_4)$.

2. Derive the Mediation Formula

Our goal is to solve for the key counterfactual quantity $\mathbb{E}[Y^{a, M^{a^*}} \mid C = c]$. We will set $a = 1$ and $a^* = 0$ for clarity, so we are solving for $\mathbb{E}[Y^{a=1, M^{a^*=0}} \mid C = c]$.

Recall the *general* mediation formula, using an integral for continuous M :

$$\mathbb{E}[Y^{1, M^0} \mid C = c] = \int_{-\infty}^{\infty} \mathbb{E}[Y \mid A = 1, M = m, C = c] p(M = m \mid A = 0, C = c) dm$$

Let's plug our regression models into this general formula.

Part 1: $\mathbb{E}[Y \mid A = 1, M = m, C = c]$ From our outcome model (Model 2), we just set $a = 1$:

$$\begin{aligned} \mathbb{E}[Y \mid A = 1, M = m, C = c] &= \theta_0 + \theta_1(1) + \theta_2 m + \theta_3' c + \theta_4(1 \times m) \\ &= (\theta_0 + \theta_1 + \theta_3' c) + (\theta_2 + \theta_4)m \end{aligned}$$

Part 2: $p(M = m \mid A = 0, C = c)$ From our mediator model (Model 1), this is simply the probability density of a Normal distribution with its mean set by $a = 0$:

$$p(m \mid A = 0, C = c) \sim \mathcal{N}(\mu = \beta_0 + \beta_1(0) + \beta_2' c, \sigma_m^2)$$

$$= \mathcal{N}(\mu = \beta_0 + \beta'_2 c, \sigma_m^2)$$

Putting it together: Now we substitute these two parts back into the integral:

$$\mathbb{E}[Y^{1,M^0} | C = c] = \int [(\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)m] p(m | A = 0, C = c) dm$$

We can split the integral and pull out terms that are constant *with respect to m*:

$$= (\theta_0 + \theta_1 + \theta'_3 c) \int p(m | A = 0, C = c) dm + (\theta_2 + \theta_4) \int m \cdot p(m | A = 0, C = c) dm$$

This integral simplifies beautifully based on two fundamental properties of probability: 1. $\int p(m | \dots) dm = 1$ (The total probability density must integrate to 1) 2. $\int m \cdot p(m | A = 0, C = c) dm = \mathbb{E}[M | A = 0, C = c]$ (This is the *definition* of the expected value of M given $A = 0$ and $C = c$)

Substituting these in:

$$\mathbb{E}[Y^{1,M^0} | C = c] = (\theta_0 + \theta_1 + \theta'_3 c) \cdot (1) + (\theta_2 + \theta_4) \cdot \mathbb{E}[M | A = 0, C = c]$$

Finally, we plug in the formula for $\mathbb{E}[M | A = 0, C = c]$ from Model 1:

$$\mathbb{E}[Y^{1,M^0} | C = c] = (\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)$$

This is the **analytical solution for the mediation formula** under these specific linear models.

3. Compute the NDE and NIE

Now we can use this result to derive the NDE and NIE. (Note: These effects are all conditional on $C = c$).

Natural Direct Effect (NDE)

$$\text{NDE} = \mathbb{E}[Y^{1,M^0} | C = c] - \mathbb{E}[Y^{0,M^0} | C = c]$$

- **First term:** We just derived this: $\mathbb{E}[Y^{1,M^0} | C = c] = (\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)$
- **Second term:** We use the same formula, but set $a = 0$ and $a^* = 0$: $\mathbb{E}[Y^{0,M^0} | C = c] = (\theta_0 + \theta_1(0) + \theta'_3 c) + (\theta_2 + \theta_4(0))(\beta_0 + \beta'_2 c) = (\theta_0 + \theta'_3 c) + \theta_2(\beta_0 + \beta'_2 c)$
- **Subtract:** $\text{NDE} = [(\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)] - [(\theta_0 + \theta'_3 c) + \theta_2(\beta_0 + \beta'_2 c)]$ $\text{NDE} = \theta_1 + \theta_4(\beta_0 + \beta'_2 c)$

Intuition for NDE: The Natural Direct Effect is *not* just θ_1 . It is θ_1 (the direct effect when $M = 0$) plus an interaction component θ_4 that is *scaled by the natural value of the mediator under control*, $\mathbb{E}[M | A = 0, C = c] = \beta_0 + \beta'_2 c$. This makes perfect sense: it's the direct effect of A , evaluated at the natural level of M that would occur if A were 0.

Natural Indirect Effect (NIE)

$$\text{NIE} = \mathbb{E}[Y^{1,M^1} | C = c] - \mathbb{E}[Y^{1,M^0} | C = c]$$

- **First term:** We use the formula, setting $a = 1$ and $a^* = 1$: $\mathbb{E}[Y^{1,M^1} | C = c] = (\theta_0 + \theta_1(1) + \theta'_3 c) + (\theta_2 + \theta_4(1))(\beta_0 + \beta_1(1) + \beta'_2 c) = (\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta_1 + \beta'_2 c)$
- **Second term:** We already have this: $\mathbb{E}[Y^{1,M^0} | C = c] = (\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)$
- **Subtract:** $\text{NIE} = [(\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta_1 + \beta'_2 c)] - [(\theta_0 + \theta_1 + \theta'_3 c) + (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)]$
The $(\theta_0 + \theta_1 + \theta'_3 c)$ terms cancel. $\text{NIE} = (\theta_2 + \theta_4)(\beta_0 + \beta_1 + \beta'_2 c) - (\theta_2 + \theta_4)(\beta_0 + \beta'_2 c)$ Factor out the common term $(\theta_2 + \theta_4)$: $\text{NIE} = (\theta_2 + \theta_4)[(\beta_0 + \beta_1 + \beta'_2 c) - (\beta_0 + \beta'_2 c)]$ $\text{NIE} = (\theta_2 + \theta_4)\beta_1$

Intuition for NIE: The Natural Indirect Effect is the product of two terms: 1. β_1 : The effect of the exposure A on the mediator M . 2. $(\theta_2 + \theta_4)$: The effect of the mediator M on the outcome Y *when the exposure is set to $A = 1$* .

This is the “product of coefficients” method, but *only* because we correctly identified which product to use: it's the $A \rightarrow M$ path multiplied by the $M \rightarrow Y$ path *under the treated condition*.

Summary

Given the two linear models: $M = \beta_0 + \beta_1 a + \beta'_2 c$ - $Y = \theta_0 + \theta_1 a + \theta_2 m + \theta'_3 c + \theta_4(a \times m)$

And assuming the four main identification assumptions hold (no unmeasured confounding, and no M-Y confounder is affected by A), the conditional natural effects are:

- $\text{NDE}(c) = \theta_1 + \theta_4(\beta_0 + \beta'_2 c)$
- $\text{NIE}(c) = \beta_1(\theta_2 + \theta_4)$

This derivation clearly shows why simply “controlling for the mediator” in a regression, as in the Baron-Kenny method, is insufficient when interaction is present. That old method would estimate the direct effect as θ_1 , which is clearly incorrect. The mediation formula, in contrast, gives us a rigorous and interpretable result.

Next Steps

This derivation gives us the *conditional* effects, which depend on the values of the confounders $C = c$. To report a single NDE and NIE for the population, we would typically average these conditional effects over the distribution of C in our population. This is the next logical step in estimation.